Fix today the rate on a future investment

LECTURE 9 · REVIEW NOTES

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Suppose our company will receive USD 1 million one year from today and invest it until year 3. I want you to locate the uncertainty before calculating anything: the money is not available now, and we do not know the two-year investment rate that will be available when it arrives. That future rate is the exposure we want to manage.



One option is to wait until year 1 and accept the two-year market rate then. The other is to agree today on a rate for that future period. Fixing 5% gives up the benefit of higher future rates in exchange for protection against lower rates.

Keep the periods distinct

$FRA_{1,3}$ is the annual rate agreed today for year 1 to year 3. $R_{1,3}$ is the matching two-year market rate observed at year 1. $R_{0,2}$ is today's two-year spot rate, applying from today to year 2. The two market rates cover equal lengths but different dates.

All rates in these notes are annual nominal rates compounded semiannually. For T years, the growth factor is (1 + rate / 2) raised to 2T. The one-year wait before investing is not part of the two-year investment period.

$$G(R_{t_1,t_2}) = \left(1 + \frac{R_{t_1,t_2}}{2}\right)^{2T}, \qquad T = t_2 - t_1$$

The payoff convention used here

We calculate an end-date payoff offsetting the difference between compounded market and locked-rate cash flows. This is the lecture's terminal-cash-flow hedge convention. Market FRA contracts commonly settle at the start of the interest period; their settlement mechanics are outside this example.

Investment: lower rates trigger a receipt

The investor wants the investment and derivative together to deliver the accumulation implied by the locked rate. Below the FRA rate, the investment earns less and the derivative pays the investor. Above the FRA rate, the investment earns more and the investor pays on the derivative.

$$\text{Payoff}_{t_2}^{\text{invest}} = N \times \left[\left(1 + \frac{\text{FRA}_{t_1,t_2}}{2} \right)^{2T} - \left(1 + \frac{R_{t_1,t_2}}{2} \right)^{2T} \right]$$

Use N = USD 1,000,000, t_1 = 1, t_2 = 3, and $\text{FRA}_{1,3}$ = 5%. There are four semiannual periods. The table separates interest from principal; a positive payoff means cash received by the investor.

Market $R_{1,3}$	Market interest	Investor payoff	Hedged interest
3%	61,363.55	42,449.34	103,812.89
5%	103,812.89	0.00	103,812.89
7%	147,523.00	-43,710.11	103,812.89

At 3%, the investment earns USD 61,363.55. The derivative pays USD 42,449.34. Their sum is USD 103,812.89, the interest earned at the locked 5% rate.

$$\text{Hedged interest} = N \left[(1 + R_{t_1,t_2}/2)^{2T} - 1 \right] + \text{Payoff}_{t_2}^{\text{invest}}$$

Including principal, the hedged terminal receipt is USD 1,103,812.89. A negative derivative payoff when rates rise offsets the extra market interest, preserving the same total.

Use the compounded difference

For this two-year period, N x (FRA - R) x T is a simple-interest approximation, not the calculation used here. Compound each rate over four half-year periods, then take the difference.

Borrowing: higher rates trigger a receipt

A company will borrow USD 1 million at year 1 and repay it at year 2. It locks 5% for that one-year period. Rising rates hurt the borrower, so the payoff has the opposite sign from the investor's payoff.

$$\text{Payoff}_{t_2}^{\text{borrow}} = N \times \left[\left(1 + \frac{R_{t_1,t_2}}{2} \right)^{2T} - \left(1 + \frac{\text{FRA}_{t_1,t_2}}{2} \right)^{2T} \right]$$

Here T = 1, so there are two semiannual periods. A positive payoff is received by the borrower and reduces borrowing cost. The loan and the FRA are separate transactions.

Market $R_{1,2}$	Loan interest	Borrower payoff	Net interest cost
3%	30,225.00	-20,400.00	50,625.00
5%	50,625.00	0.00	50,625.00
7%	71,225.00	20,600.00	50,625.00

At 7%, loan interest is USD 71,225. The FRA pays USD 20,600, leaving USD 50,625 of net interest cost. At 3%, interest is only USD 30,225, but the borrower pays USD 20,400 on the FRA. The combined cost is again USD 50,625.

$$\text{Net interest cost} = N \left[(1 + R_{t_1,t_2}/2)^{2T} - 1 \right] - \text{Payoff}_{t_2}^{\text{borrow}}$$

Compare matching maturities

At year 1, compare $\text{FRA}_{1,2}$ with the one-year market rate $R_{1,2}$. A six-month rate or today's one-year rate applies to a different exposure. Cancellation requires the notional, timing and rate convention to match the underlying cash flow.

Read the sign from the exposure

Future investor: rates down are bad, so receive when rates fall. Future borrower: rates up are bad, so receive when rates rise. Then check that the combined interest amount is constant.

Derive the forward rate from today's curve

Investing today until t_2 should produce the same accumulation as investing until t_1 and locking the forward investment from t_1 to t_2. The two routes produce the same dated cash flow, so their growth factors must agree.



$$\left(1+\frac{R_{0,t_2}}{2}\right)^{2t_2}=\left(1+\frac{R_{0,t_1}}{2}\right)^{2t_1}\left(1+\frac{\mathrm{FRA}_{t_1,t_2}}{2}\right)^{2(t_2-t_1)}$$

$$\mathrm{FRA}_{t_1,t_2}=2\left[\left(\frac{(1+R_{0,t_2}/2)^{2t_2}}{(1+R_{0,t_1}/2)^{2t_1}}\right)^{\frac{1}{2(t_2-t_1)}}-1\right]$$

For a separate pricing example, today's one-year zero rate is 4% and its three-year zero rate is 5%. The forward period, year 1 to year 3, contains four semiannual periods.

$$\mathrm{FRA}_{1,3}=2\left[\left(\frac{1.025^6}{1.02^2}\right)^{1/4}-1\right]\approx 5.5018\%$$

The result is 5.5018% annually, compounded semiannually. It differs from the assumed 5% contract on the preceding pages because we are pricing a new forward from a specified spot curve. Once locked, use this rate in the same payoff equations.

An implied rate is not a forecast

This calculation uses prices available today. It does not require the future market rate $R_{1,3}$ to equal the forward. A different realized market rate produces the offsetting derivative payoff.